

Chapter 8 Study Sheet

Similarity — everything you need on one page.

8.1 Similar Polygons

- Similar ($\sim$): corresponding angles congruent AND corresponding sides proportional.
- Scale factor = the shared ratio of corresponding sides. It has a direction: A to B is the reciprocal of B to A.
- The similarity statement's letter order IS the correspondence — read matching parts by position.
- Perimeters of similar figures share the scale factor.
- All squares are similar; all equilateral triangles are similar; rectangles only sometimes.

8.2-8.3 Proving Triangles Similar

- AA: two pairs of congruent angles. (The third pair is free, by the Triangle Sum Theorem.)
- SSS Similarity: all THREE side ratios equal.
- SAS Similarity: two side ratios equal and the INCLUDED angles congruent.
- Proportional sides with a non-included congruent angle prove nothing — the SSA trap, similarity edition.
- Parallel lines are AA machines: alternate interior angles and vertical angles hand you the two angle pairs.
- Indirect measurement: height/height = shadow/shadow at the same moment (AA via the sun).

8.4 Proportionality Theorems

- Triangle Proportionality: a line parallel to one side divides the other two sides proportionally — and conversely.
- Pieces with pieces, wholes with wholes: $BX/XA = BY/YC$, or $BX/BA = BY/BC$. Never one of each.
- Three parallel lines divide any two transversals proportionally.
- Angle bisector: splits the opposite side in the ratio of the two adjacent sides. Solve with pieces $5k$ and $4k$, etc.